# Reliable and Large-Scale Computer System (RELACS) Research LAB

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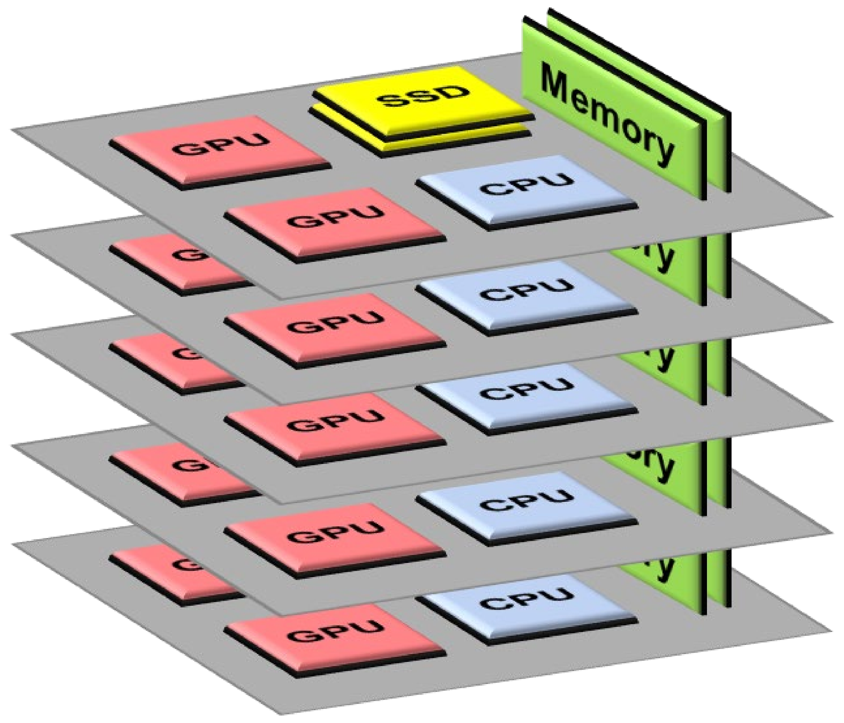
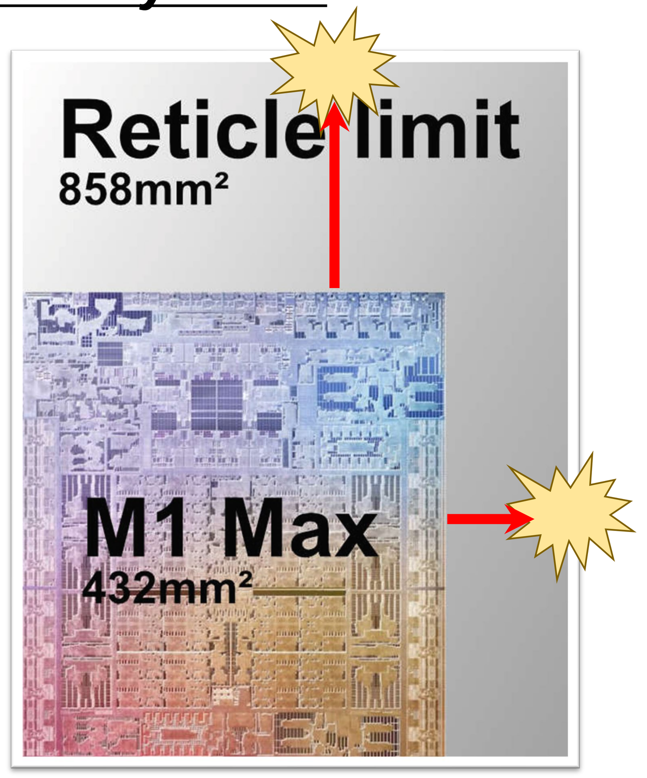
## Background: Tougher Resource Expansion

- Resource expansion is becoming tougher and tougher due to form-factor limits nowadays…
  - **System-level?** Few slots on motherboard & saturated pins
  - **Chip-level?** Transistors on a die is reaching reticle limit
    - Reticle limit: max. area with a single lithography shot
- ***Question : How to overcome scalability limit?***

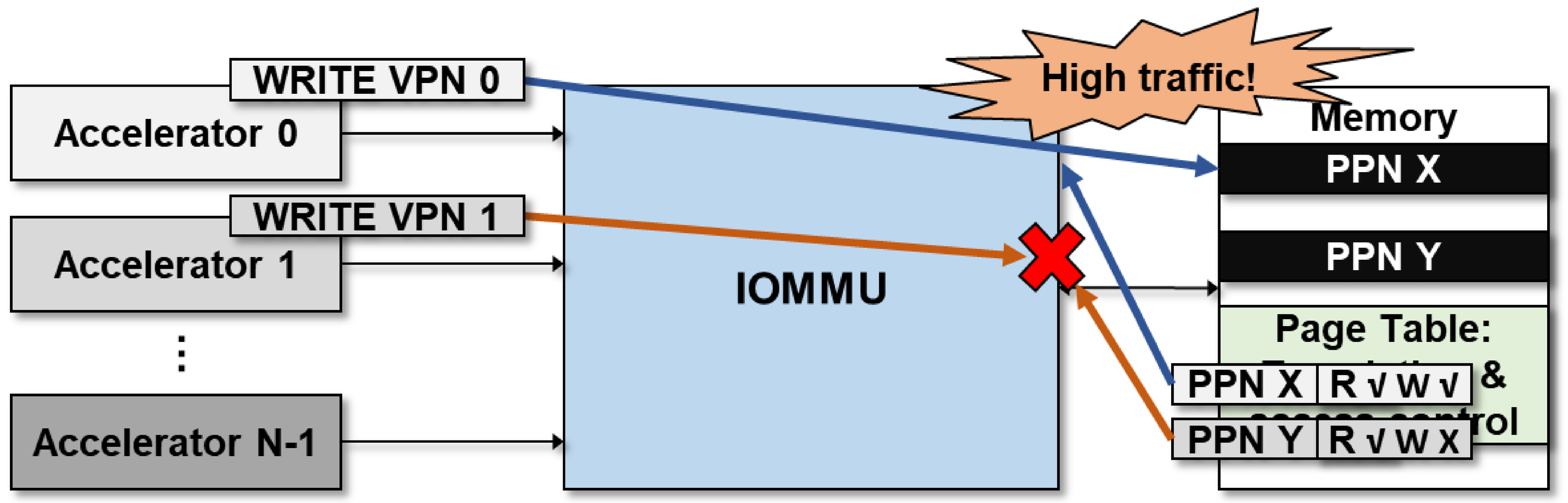



## Topic 1: Security—The Red Line of Systems

### Sandboxing from Malicious Third-Party IPs

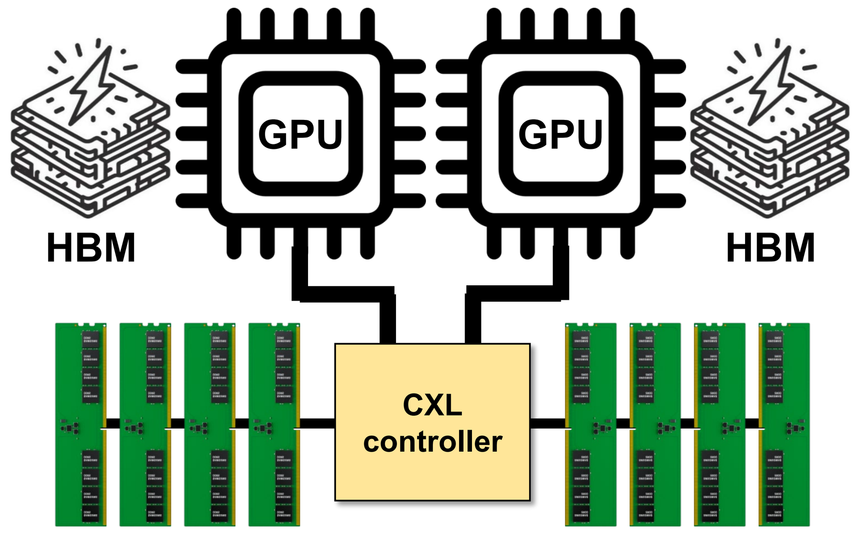
- Develop secure architecture while disaggregating accelerators
- Mission 1: Sandbox from malicious accelerators
  - Address translation is common in accelerator-rich systems
  - Accelerators may manipulate translation & permission info.
- Mission 2: Sandbox from malicious IOMMU
  - IOMMU conducts address translation
  - IOMMU can be 3rd party in highly disaggregated systems

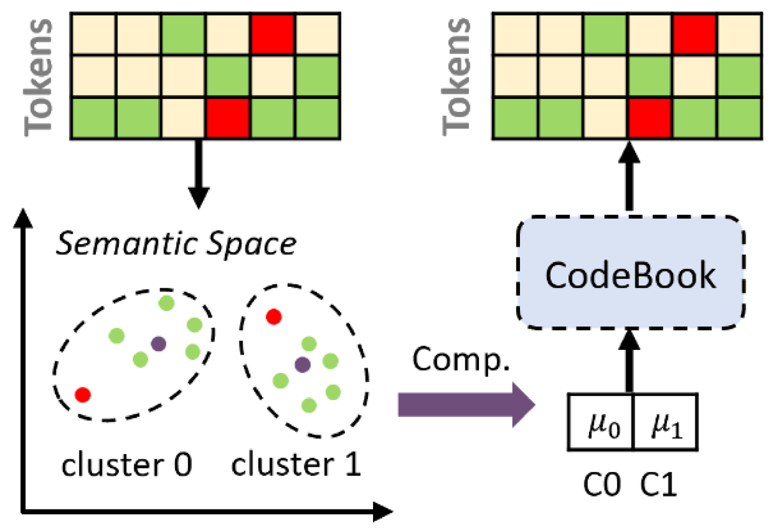



## Topic 2: Compression—Towards High Availability

### Lossy Compression for Large-Language Model (LLM)

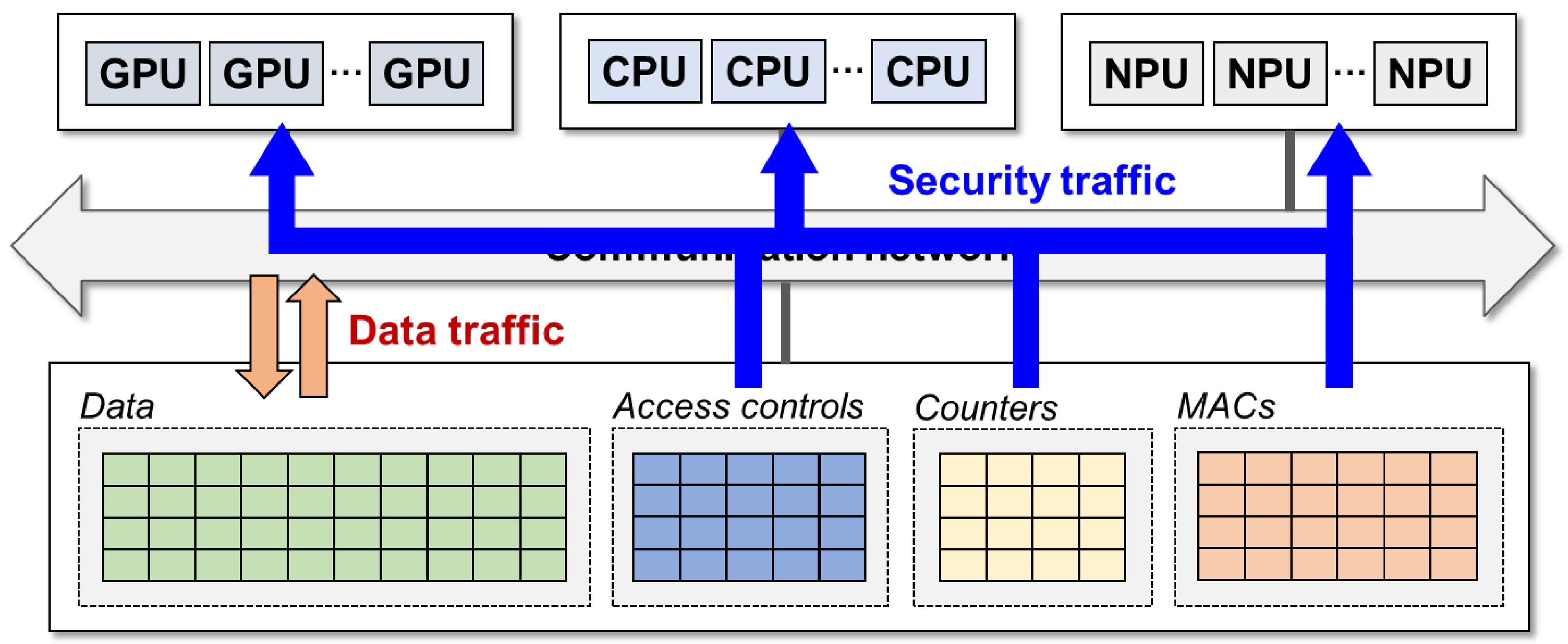
- KV cache of modern LLM inference is a troublesome
  - KV Cache? Context information representing history
  - More tokens, larger KV cache → larger memory is needed
- CXL memory is a key to accommodate large KV cache with high cost-efficiency compared to adding more GPUs/NPUs
- ***Challenge: CXL memory is much slower than HBMs***
  - Think about their bandwidth gap…



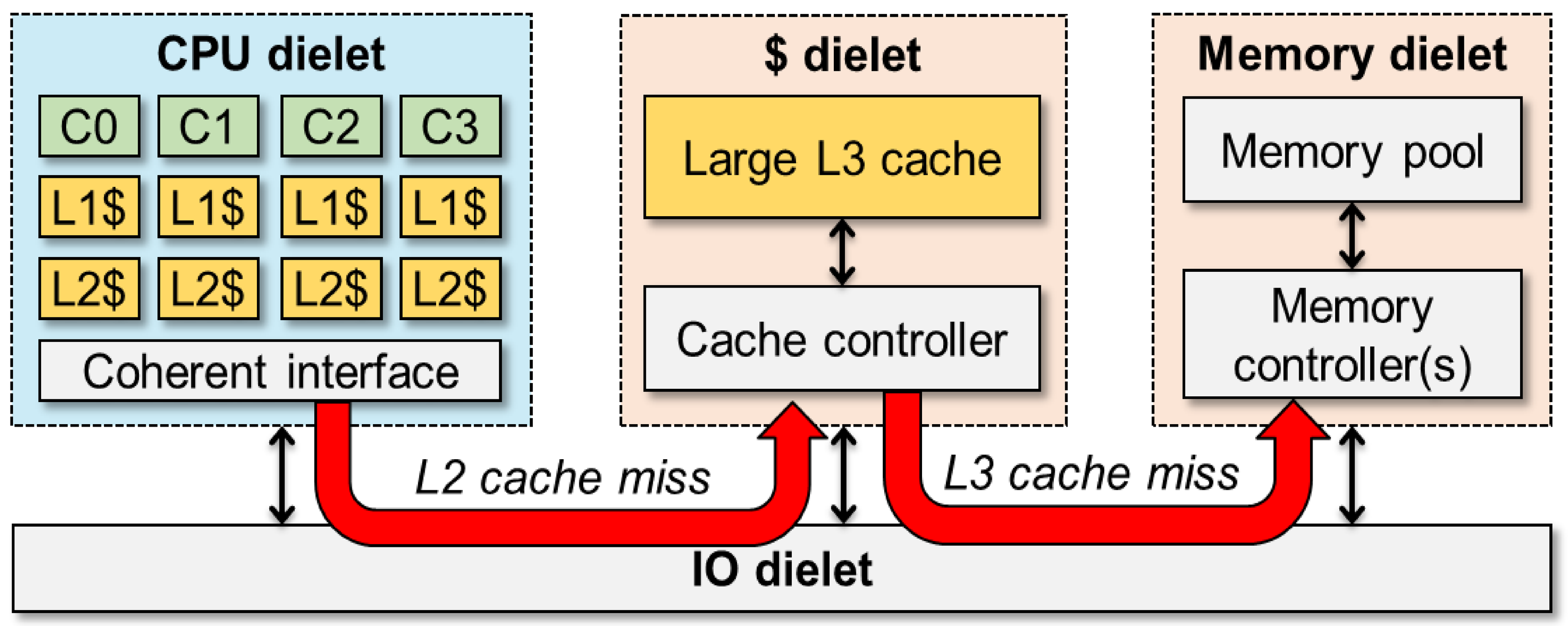

## Road to Future: Resource Disaggregation!

- Alternative architecture? ***Resource disaggregation***!
  - Aggregate number of similar components
  - Defined either at chip-level (chiplet) or system-level
  - ***Goal*** Higher scalability, availability, and robustness
- ***Fundamental problem: Traffic congested on network!***
  - Metadata traffic: security & compression metadata…
  - Data traffic: user- or application-level data



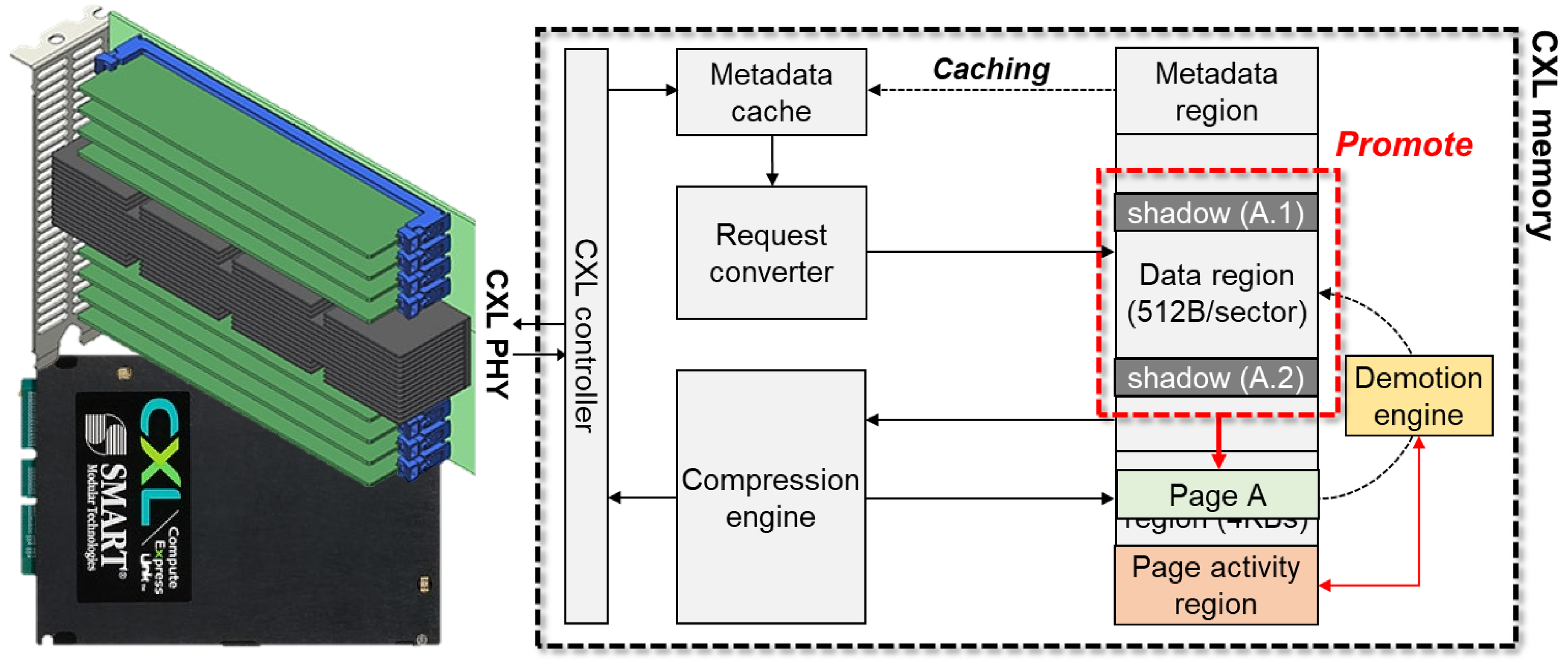

### Sandboxing from Malicious Memory Controller (MC)

- Why 3rd party MC? What's wrong with it?
  - Chiplet may import MC with complicated policy as 3rd party
  - MC could be tape-out dielet → Memory encryption engine (MEE) now shouldn't be in MC
- Potential architectural issues of decoupling MEE from MC
  - ***Latency exposure*** due to faster data fetch from L3 cache
  - ***Cache pollution on L3 cache*** would be problematic for workloads having huge working set




### Lossless, Form-Factor-Aware Compression for CXL

- CXL is considered a candidate for memory disaggregation
- BUT its form factor limit entails several ***practical challenges***
  - Fewer memory devices than expected
    → Low capacity & bandwidth
  - ***Memory compression*** should be in CXL memory
- ***Challenge: Compression, especially block level, incurs bulky data movements within CXL memory***




## · Group Information & Achievements

- Members Now: 2 Ph.D. candidates 2 Integrated + M.S.-Ph.D. course (석박통합과정) students
**** Recruiting Integrated M.S.-Ph.D. course! 석박통합과정 모집중!***
- Published at 9 top-tier conference venues (MICRO, HPCA, PACT, ICS…), 8 top-tier journals

## · Benefits

- Close mentorship with exempted from tuition fee (등록금)
- Attending international conferences
- Accessible to valuable infrastructure